Complete cold DLP runtime through ecbench

Run 20261019-165520-boolean-recursive-ecbenc-9d61aa, source 0acfc55b7f. Complete one-target cold DLP recovery on dedicated c7i.2xlarge; native ecbench L2 requirement, one worker,8 independent public targets per curve,5 measured rounds plus1 warmup,11 interleaved arms. Sources/specs and complete audit certificates remain adjacent/raw. Missing, failed and below-level rows are retained; a launch is not a result.

All methods within each session share the same public targets and standard polynomial subspace: dimension5 at n17 and6 at n23. These bases differ from the component experiment's Frobenius-invariant bases. The curve identities are icv1-f2m17-tm101-00378d4e and icv1-f2m23-tm5197-1f85e9e1. This two-size panel is not a scaling claim.

The pipeline solve clock includes factor-base construction, oracle prepare, relation collection, linear algebra, target descent/recovery and internal verification. Whole child process additionally includes adapter construction, report metadata, common workload construction and startup. Both clocks are recorded; the native harness's comparison gate uses the pipeline solve clock. Independent parent verification/ replay, compilation and warmup are outside these clocks.

F4/F5 baseline uses the prior recursive policy with qualified GF2 kernels. Sources0acfc55b7f/71c4c99d85 use candidate matrix_runtime=quadratic-tail; corrected source9bed104b0a uses adaptive-quadratic-tail, with a different method identity. F6 names kernel_runtime=native-v1. Generalized typed roots and both Semaev table backends participate. Environment-override refusal remains intact; no admission threshold is relaxed.

Wall ratios below are B/A from ecbench's own paired bootstrap and A/A gate. Admission, completeness and outside_noise must all support a gain; partial/failed sessions do not become successful comparisons. Process medians are supplementary observed clocks, not a replacement gate.

Counted S, generic floor and matched strong rho are retained in table-n*.txt and ecbench-summary.json. Gray assignments are explicitly counted unpriced; reference native unpriced counters are retained too. Sensitivity bands at0/0.1/1 GAE per recorded operation are diagnostic and deduplicate generalized solver-count aliases. An unquantified counter stays unquantified. No ratio of lower bounds is a full-cost ranking.

Exact original-output and replay checks, query-count comparisons, source/binary hashes, failures and job resources remain in the evidence. All-L2 coverage is separate from execution/verification completeness; a fully verified session can have below-level timing rows. Low-sample clocks are descriptive; no threshold is relaxed. New matrix policies stay opt-in; no external-fastest, other-hardware or rho-beating claim is made.

n17: 528 total records, 88 warmups; replay audit OK True; all requested L2 timing coverage False.

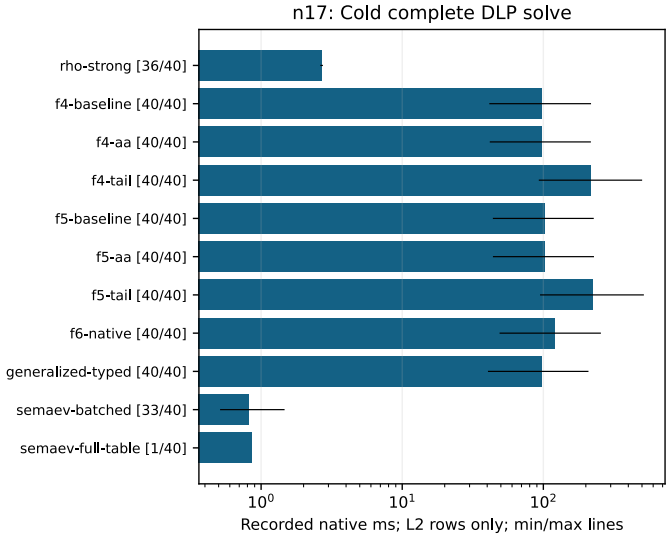
n17 f4: candidate/baseline pipeline wall 2.2673, CI95 [2.228161831919641, 2.2774882139987653], gate admitted, outside A/A noise True. Ratios above1 are regressions, below1 are reductions in time; admission/completeness still govern interpretation.

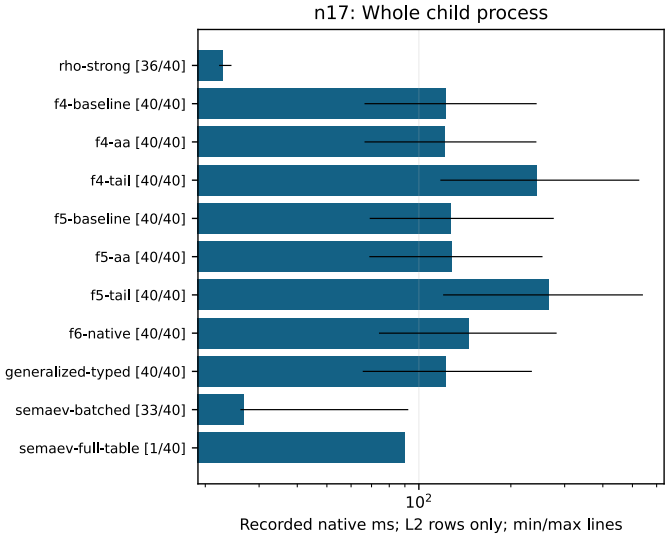
n17 f5: candidate/baseline pipeline wall 2.1947, CI95 [2.1672102399596103, 2.223400789544616], gate admitted, outside A/A noise True. Ratios above1 are regressions, below1 are reductions in time; admission/completeness still govern interpretation.

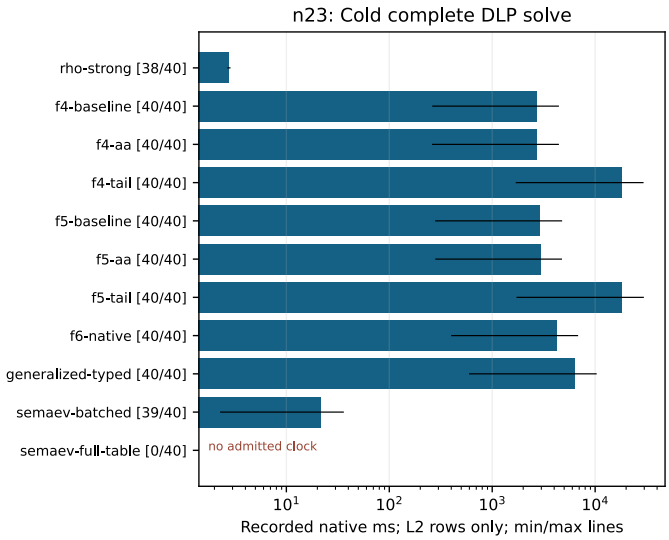
n23: 528 total records, 88 warmups; replay audit OK True; all requested L2 timing coverage False.

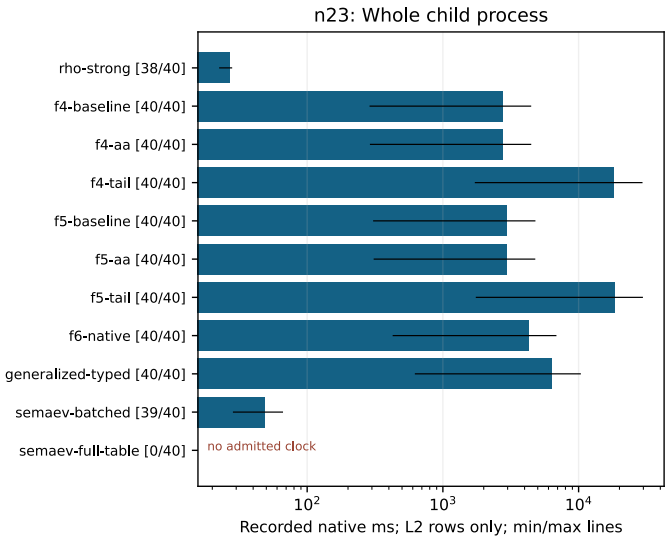
n23 f4: candidate/baseline pipeline wall 6.5737, CI95 [6.56333553393357, 6.591094514060304], gate admitted, outside A/A noise True. Ratios above1 are regressions, below1 are reductions in time; admission/completeness still govern interpretation.

n23 f5: candidate/baseline pipeline wall 6.2148, CI95 [6.202408767259975, 6.225611004921411], gate admitted, outside A/A noise True. Ratios above1 are regressions, below1 are reductions in time; admission/completeness still govern interpretation.


n17: Cold complete DLP solve
rho-strong [36/40]
f4-baseline [40/40]
f4-aa [40/40]
f4-tail [40/40]
f5-baseline [40/40]
f5-aa [40/40]
f5-tail [40/40]
f6-native [40/40]
generalized-typed [40/40]
semaev-batched [33/40]
semaev-full-table [1/40]
10^0
10^1
10^2
Recorded native ms; L2 rows only; min/max lines


n17: Whole child process
rho-strong [36/40]
f4-baseline [40/40]
f4-aa [40/40]
f4-tail [40/40]
f5-baseline [40/40]
f5-aa [40/40]
f5-tail [40/40]
f6-native [40/40]
generalized-typed [40/40]
semaev-batched [33/40]
semaev-full-table [1/40]
10^2
Recorded native ms; L2 rows only; min/max lines


n23: Cold complete DLP solve
rho-strong [38/40]
f4-baseline [40/40]
f4-aa [40/40]
f4-tail [40/40]
f5-baseline [40/40]
f5-aa [40/40]
f5-tail [40/40]
f6-native [40/40]
generalized-typed [40/40]
semaev-batched [39/40]
semaev-full-table [0/40]
no admitted clock
10^1
10^2
10^3
10^4
Recorded native ms; L2 rows only; min/max lines


n23: Whole child process
rho-strong [38/40]
f4-baseline [40/40]
f4-aa [40/40]
f4-tail [40/40]
f5-baseline [40/40]
f5-aa [40/40]
f5-tail [40/40]
f6-native [40/40]
generalized-typed [40/40]
semaev-batched [39/40]
semaev-full-table [0/40]
no admitted clock
10^2
10^3
10^4
Recorded native ms; L2 rows only; min/max lines

Recorded clocks and admission

n	Arm	Solve ms	Process ms	Verified	L2	Requested	All L2
17	rho-strong	2.6638	22.7818	40	36	40	False
17	f4-baseline	96.8280	122.1154	40	40	40	True
17	f4-aa	96.4936	121.8268	40	40	40	True
17	f4-tail	216.7685	242.0721	40	40	40	True
17	f5-baseline	101.7174	126.7612	40	40	40	True
17	f5-aa	102.1956	127.8513	40	40	40	True
17	f5-tail	222.9573	266.7999	40	40	40	True
17	f6-native	119.9306	145.0248	40	40	40	True
17	generalized-typed	96.9606	122.1977	40	40	40	True
17	semaev-batched	0.8113	26.6584	40	33	40	False
17	semaev-full-table	0.8623	89.5996	40	1	40	False
23	rho-strong	2.7651	26.7752	40	38	40	False
23	f4-baseline	2736.7244	2764.5258	40	40	40	True
23	f4-aa	2735.7468	2765.9796	40	40	40	True
23	f4-tail	17981.2337	18011.3930	40	40	40	True
23	f5-baseline	2933.7356	2963.8713	40	40	40	True
23	f5-aa	2937.0158	2967.3401	40	40	40	True
23	f5-tail	18262.5992	18293.1179	40	40	40	True
23	f6-native	4210.5229	4240.7998	40	40	40	True
23	generalized-typed	6285.0703	6310.7315	40	40	40	True
23	semaev-batched	21.6131	48.7818	40	39	40	False
23	semaev-full-table	missing	missing	40	0	40	False

Native harness comparisons; B/A

n	Pair	Wall gate	Median ratio	CI95	Outside noise	Below L2
17	f4	admitted	2.2673	2.2282..2.2775	True	0
17	f5	admitted	2.1947	2.1672..2.2234	True	0
17	rho-f4-tail	admitted	86.5500	75.9717..101.2833	True	4
17	rho-f5-tail	admitted	88.7481	77.3244..103.9905	True	4
17	rho-f6-native	admitted	47.6281	40.2044..52.3683	True	4
17	rho-generalized-typed	admitted	38.7422	32.8392..42.2406	True	4
17	rho-semaev-batched	admitted	0.3090	0.2891..0.3327	True	10
17	rho-semaev-full-table	descriptive	0.4035	missing	None	39
23	f4	admitted	6.5737	6.5633..6.5911	True	0
23	f5	admitted	6.2148	6.2024..6.2256	True	0
23	rho-f4-tail	admitted	6380.3279	4677.1999..7380.0729	True	2
23	rho-f5-tail	admitted	6480.1291	4739.9090..7495.2008	True	2
23	rho-f6-native	admitted	1494.0203	1095.4924..1729.1124	True	2
23	rho-generalized-typed	admitted	2230.1848	1658.1652..2585.5553	True	2
23	rho-semaev-batched	admitted	7.6036	6.5636..8.9156	True	3
23	rho-semaev-full-table	descriptive	7.7986	missing	None	40

Partly priced S sensitivity; all entries diagnostic

n	Arm	0 GAE/op	0.1 GAE/op	1 GAE/op	Lower bound	Unknown entries
17	rho-strong	4.0416	4.185	5.4753	True	0
17	f4-baseline	43.799	43.799	43.799	False	0
17	f4-aa	43.799	43.799	43.799	False	0
17	f4-tail	143.67	143.74	144.29	True	0
17	f5-baseline	43.731	43.731	43.731	False	0
17	f5-aa	43.731	43.731	43.731	False	0
17	f5-tail	142.99	143.05	143.6	True	0
17	f6-native	33.082	33.082	33.082	False	0
17	generalized-typed	16.421	1080.1	10653	True	0
17	semaev-batched	17.317	17.317	17.317	False	0
17	semaev-full-table	18.083	18.083	18.083	False	0
23	rho-strong	0.88552	0.95031	1.5335	True	0
23	f4-baseline	72.251	72.251	72.251	False	0
23	f4-aa	72.251	72.251	72.251	False	0
23	f4-tail	1138.9	1138.9	1139.2	True	0
23	f5-baseline	72.228	72.228	72.228	False	0
23	f5-aa	72.228	72.228	72.228	False	0
23	f5-tail	1137.1	1137.2	1137.5	True	0
23	f6-native	65.541	65.541	65.541	False	0
23	generalized-typed	43.007	16897	1.6859e+05	True	0
23	semaev-batched	43.709	43.709	43.709	False	0
23	semaev-full-table	44.039	44.039	44.039	False	0

Explicit unpriced operation ledgers; different units are not equated

n	Arm	Unit	Samples	Median count	Min	Max
17	rho-strong	canonicalisations_uncharged	40	192.0	64	356
17	rho-strong	partition_hashes_uncharged	40	148.5	30	306
17	rho-strong	table_inserts_uncharged	40	9.0	2	20
17	rho-strong	table_queries_uncharged	40	10.0	3	22
17	f4-tail	quadratic_tail_assignments_uncharged	40	132.0	4	426
17	f5-tail	quadratic_tail_assignments_uncharged	40	132.0	4	426
17	generalized-typed	generalized_assignments_uncharged	40	2613760.0	1051648	5709824
23	rho-strong	canonicalisations_uncharged	40	672.0	160	1217
23	rho-strong	partition_hashes_uncharged	40	594.5	119	1105
23	rho-strong	table_inserts_uncharged	40	39.5	12	79
23	rho-strong	table_queries_uncharged	40	40.5	13	80
23	f4-tail	quadratic_tail_assignments_uncharged	40	599.0	12	2000
23	f5-tail	quadratic_tail_assignments_uncharged	40	599.0	12	2000
23	generalized-typed	generalized_assignments_uncharged	40	355270656.0	33546240	580587520

Admitted phase clocks; recorded ms

n	Arm	Phase	Median ms	Min ms	Max ms	Samples
17	f4-baseline	factor_base	0.1379	0.1318	0.1626	40
17	f4-baseline	linear_algebra	0.0093	0.0023	0.0142	40
17	f4-baseline	oracle_setup	0.0430	0.0398	0.0630	40
17	f4-baseline	relations	96.6016	41.2053	217.4618	40
17	f4-baseline	verify	0.0020	0.0016	0.0025	40
17	f4-aa	factor_base	0.1369	0.1334	0.1625	40
17	f4-aa	linear_algebra	0.0093	0.0023	0.0140	40
17	f4-aa	oracle_setup	0.0427	0.0405	0.0483	40
17	f4-aa	relations	96.2727	41.5140	216.9719	40
17	f4-aa	verify	0.0021	0.0016	0.0033	40
17	f4-tail	factor_base	0.1373	0.1306	0.1563	40
17	f4-tail	linear_algebra	0.0097	0.0025	0.0133	40
17	f4-tail	oracle_setup	0.0425	0.0402	0.0626	40
17	f4-tail	relations	216.5320	92.6474	500.9037	40
17	f4-tail	verify	0.0020	0.0017	0.0027	40
17	f5-baseline	factor_base	0.1383	0.1308	0.1678	40
17	f5-baseline	linear_algebra	0.0092	0.0026	0.0168	40
17	f5-baseline	oracle_setup	0.0427	0.0396	0.0485	40
17	f5-baseline	relations	101.4892	43.6727	227.3611	40
17	f5-baseline	verify	0.0020	0.0016	0.0024	40
17	f5-aa	factor_base	0.1372	0.1316	0.1512	40
17	f5-aa	linear_algebra	0.0094	0.0026	0.0130	40
17	f5-aa	oracle_setup	0.0419	0.0395	0.0478	40
17	f5-aa	relations	101.9722	43.7178	228.1939	40
17	f5-aa	verify	0.0020	0.0017	0.0026	40
17	f5-tail	factor_base	0.1384	0.1310	0.1535	40
17	f5-tail	linear_algebra	0.0093	0.0023	0.0153	40
17	f5-tail	oracle_setup	0.0421	0.0394	0.0573	40
17	f5-tail	relations	222.7332	94.5572	515.3173	40
17	f5-tail	verify	0.0021	0.0017	0.0048	40
17	f6-native	factor_base	0.1360	0.1311	0.1615	40
17	f6-native	linear_algebra	0.0096	0.0025	0.0142	40
17	f6-native	oracle_setup	0.0425	0.0393	0.0619	40
17	f6-native	relations	119.7100	48.7445	255.4314	40
17	f6-native	verify	0.0020	0.0016	0.0026	40
17	generalized-typed	factor_base	0.1377	0.1312	0.1515	40
17	generalized-typed	linear_algebra	0.0089	0.0027	0.0211	40
17	generalized-typed	oracle_setup	0.0417	0.0391	0.0470	40
17	generalized-typed	relations	96.7393	40.2845	208.5688	40
17	generalized-typed	verify	0.0020	0.0017	0.0024	40
17	semaev-batched	factor_base	0.1447	0.1388	0.1754	33
17	semaev-batched	linear_algebra	0.0063	0.0022	0.0081	33
17	semaev-batched	oracle_setup	0.0148	0.0132	0.0182	33
17	semaev-batched	relations	0.6123	0.3125	1.2736	33
17	semaev-batched	verify	0.0021	0.0017	0.0030	33
17	semaev-full-table	factor_base	0.1364	0.1364	0.1364	1
17	semaev-full-table	linear_algebra	0.0053	0.0053	0.0053	1
17	semaev-full-table	oracle_setup	0.2175	0.2175	0.2175	1
17	semaev-full-table	relations	0.4713	0.4713	0.4713	1
17	semaev-full-table	verify	0.0017	0.0017	0.0017	1
23	f4-baseline	factor_base	0.1559	0.0555	0.1698	40
23	f4-baseline	linear_algebra	0.0205	0.0033	0.0304	40
23	f4-baseline	oracle_setup	0.0736	0.0700	0.0915	40
23	f4-baseline	relations	2736.5140	260.9896	4447.7940	40
23	f4-baseline	verify	0.0040	0.0034	0.0049	40
23	f4-aa	factor_base	0.1572	0.0587	0.1748	40
23	f4-aa	linear_algebra	0.0200	0.0034	0.0352	40
23	f4-aa	oracle_setup	0.0749	0.0714	0.0820	40
23	f4-aa	relations	2735.5267	259.8684	4446.8064	40
23	f4-aa	verify	0.0040	0.0034	0.0102	40
23	f4-tail	factor_base	0.1570	0.0610	0.1901	40
23	f4-tail	linear_algebra	0.0296	0.0037	0.0487	40
23	f4-tail	oracle_setup	0.0754	0.0701	0.0941	40
23	f4-tail	relations	17980.9470	1692.1750	29546.7284	40
23	f4-tail	verify	0.0039	0.0034	0.0058	40
23	f5-baseline	factor_base	0.1556	0.0570	0.1873	40

n	Arm	Phase	Median ms	Min ms	Max ms	Samples
23	f5-baseline	linear_algebra	0.0207	0.0031	0.0297	40
23	f5-baseline	oracle_setup	0.0739	0.0704	0.0933	40
23	f5-baseline	relations	2933.5276	278.8703	4776.1124	40
23	f5-baseline	verify	0.0040	0.0035	0.0051	40
23	f5-aa	factor_base	0.1561	0.0583	0.1803	40
23	f5-aa	linear_algebra	0.0198	0.0052	0.0392	40
23	f5-aa	oracle_setup	0.0745	0.0718	0.0834	40
23	f5-aa	relations	2936.7961	279.3681	4765.4594	40
23	f5-aa	verify	0.0040	0.0034	0.0047	40
23	f5-tail	factor_base	0.1559	0.0603	0.2443	40
23	f5-tail	linear_algebra	0.0297	0.0047	0.0469	40
23	f5-tail	oracle_setup	0.0735	0.0704	0.0851	40
23	f5-tail	relations	18262.3150	1720.3922	29746.0744	40
23	f5-tail	verify	0.0040	0.0033	0.0056	40
23	f6-native	factor_base	0.1561	0.0597	0.1836	40
23	f6-native	linear_algebra	0.0220	0.0033	0.0346	40
23	f6-native	oracle_setup	0.0747	0.0697	0.0836	40
23	f6-native	relations	4210.3045	399.5331	6834.3177	40
23	f6-native	verify	0.0040	0.0035	0.0053	40
23	generalized-typed	factor_base	0.1557	0.0561	0.1788	40
23	generalized-typed	linear_algebra	0.0221	0.0031	0.0336	40
23	generalized-typed	oracle_setup	0.0725	0.0677	0.0855	40
23	generalized-typed	relations	6284.8052	595.9769	10358.3040	40
23	generalized-typed	verify	0.0040	0.0034	0.0049	40
23	semaev-batched	factor_base	0.1636	0.1586	0.1936	39
23	semaev-batched	linear_algebra	0.0151	0.0027	0.0230	39
23	semaev-batched	oracle_setup	0.0287	0.0267	0.0406	39
23	semaev-batched	relations	21.3609	2.0463	35.8618	39
23	semaev-batched	verify	0.0040	0.0034	0.0053	39